SYM Forklifts

Interactive Forklift E-Commerce & Product Configuration Platform

Role	Full Stack MERN Developer
Project Type	Production Website & Interactive Product Platform
Industry	Industrial Machinery / Forklifts
Client	Australian Forklift & Industrial Machinery Business

Frontend	React.js, JavaScript, Responsive UI
Backend	Node.js, Express.js, REST APIs
Database	MongoDB, Mongoose
Payments	Stripe
Integrations	Google Maps API, Meta Pixel, Email, SMS
AI	OpenAI, OpenClaw, AI Agents
Infrastructure	DigitalOcean, Nginx, PM2, GitHub Actions, CI/CD

Project Overview

SYM Forklifts is a production-grade Australian forklift and industrial machinery platform designed to provide customers with a modern digital purchasing experience.

Rather than functioning as a traditional catalogue website, the platform provides an **interactive machine discovery and configuration experience**, allowing customers to explore forklift models, select compatible attachments and accessories, configure their machine, view visual changes, and progress toward enquiry or purchase.

A major focus of the platform was creating a **dynamic product configuration system** where machine compatibility rules control what customers can select.

The experience was designed around the idea that customers should be able to configure a forklift visually and logically, similar to the configuration experience used by modern automotive platforms such as Tesla.

The Challenge

Industrial machinery websites often present products as static pages containing specifications, images, and enquiry forms. For SYM Forklifts, the objective was to create a much more interactive experience.

Customers needed to be able to:

- Discover different forklift models
- Understand machine specifications
- Select different configurations
- Select attachments
- Automatically determine attachment compatibility
- See unavailable combinations immediately
- Dynamically update the machine configuration
- See visual changes when an attachment is selected
- Understand which accessories work with a specific machine
- Complete the purchasing/enquiry journey
- Make secure online payments where applicable

The biggest challenge was therefore not simply displaying products, but building a **rules-driven product configuration system**.

Website Architecture

The website was developed as a complete full-stack application.

```
                SYM FORKLIFTS WEBSITE
                        |
         +--------------+--------------+
         |                             |
   Product Discovery          Product Configuration
         |                             |
  Machine Categories           Machine Selection
         |                             |
   Product Details               Attachments
         |                             |
   Specifications             Compatibility Rules
         |                             |
         +--------------+--------------+
                        |
                   REST APIs
                        |
                Node.js / Express
                        |
                     MongoDB
                        |
             Product / Compatibility
                  Configuration Data
```

The frontend communicates with backend APIs to retrieve products, specifications, configurations, attachments, compatibility information, and other dynamic content.

1. Modern Industrial Website Experience

The website was designed to move away from the traditional industrial machinery website model. Instead of presenting customers with large amounts of technical information at once, the interface focuses on:

- Clear product discovery

- Strong visual presentation
- Product-focused layouts
- Structured specifications
- Interactive configuration
- Clear calls to action
- Responsive design
- Conversion-focused customer journeys

The website combines the technical nature of forklift products with a modern consumer-style digital experience.

2. Product Discovery

A core part of the website is the machinery discovery experience. Customers can explore forklift products and understand the differences between available models before configuring a machine. Product pages are structured around information such as:

- Machine name
- Machine category
- Product imagery
- Specifications
- Features
- Available attachments
- Configuration options
- Related information
- Enquiry / purchasing actions

The objective was to make industrial machinery easier to explore without requiring customers to understand every technical specification before beginning.

3. Dynamic Product Configuration

One of the most important features developed for the platform is the **interactive product configurator**. The configurator allows customers to select options for their machine while the system dynamically determines which combinations are valid. Instead of treating every attachment as independently selectable, the platform understands relationships between machine, attachment, and compatibility.

Machine → Attachment → Compatibility

This creates a much more intelligent configuration experience.

4. Attachment Compatibility Engine

The compatibility system is one of the most technically important parts of the website. Each machine can have a defined set of compatible attachments — for example, machine QWF35 might allow a bucket, clamp, and fork positioner while other attachments remain unavailable. The frontend does not simply show every attachment as selectable; instead, the system evaluates the compatibility rules against the currently selected machine.

5. Dynamic Compatibility Rules

The compatibility rules dynamically control the available selections. When a machine is selected, the system checks compatibility and marks each attachment as available or disabled accordingly, preventing customers from creating technically invalid configurations.

6. Automatic Option Disablement

A particularly important UX rule is that selecting one attachment can immediately change the availability of other attachments. For example, if bucket, clamp, and fork positioner are all initially available and the customer selects the bucket, the system immediately marks the bucket as selected and the clamp and fork positioner as unavailable, since the current configuration no longer supports those combinations. This prevents invalid combinations from being submitted.

7. Compatibility-Based State Management

The frontend maintains the selected machine and selected configuration state. A simplified configuration flow is:

```
Select Machine
      |
Load Compatible Attachments
      |
Select Attachment
      |
Recalculate Compatibility
      |
Disable Invalid Options
      |
Update Configuration
      |
Update Visual Representation
```

This means the interface is not just visually interactive — the interaction is connected to real business rules.

8. Tesla-Style Visual Configuration

Another major feature is the **visual configuration experience**. The objective was to provide customers with immediate visual feedback when changing machine options — for example, when the customer selects a bucket attachment, the machine's visual representation updates to reflect the new configuration.

9. Dynamic Attachment Visualization

The same concept applies to different attachments. The default machine view updates to show the bucket when it's selected, or the clamp when that option is chosen instead. The actual product imagery/configuration changes according to the selected attachment, giving customers immediate visual confirmation of what they are configuring.

10. Configuration Without Page Reloads

The configuration experience was designed to feel like a modern application rather than a traditional website. When customers change options:

- The UI updates immediately

- Compatibility is recalculated
- Invalid options are disabled
- Selected options are reflected in the configuration
- Product visuals can change
- The customer remains on the same page

This creates a much smoother purchasing experience.

11. Product Configuration Data Model

The backend stores product and compatibility information in MongoDB. Conceptually, the system works around relationships where each machine has specifications, images, features, and a set of compatible attachments (bucket, clamp, fork positioner, and others). This makes the compatibility system configurable from backend data rather than hardcoding every combination directly into the frontend.

12. Product Specifications

Product pages provide structured technical information so customers can understand the machine before configuring it, including:

- Capacity
- Dimensions
- Performance information
- Machine specifications
- Features
- Attachments
- Configuration options

The information architecture was designed to make technical product data easier to scan and understand.

13. Product Image System

Visual presentation is particularly important for industrial machinery. The website uses product imagery throughout the customer journey, including product listings, product details, configuration interfaces, attachment selections, and machine previews. The objective is to make the product experience visual rather than relying only on technical text.

14. Configuration-Based Product Images

The image system is connected to configuration selections: as the customer moves from machine selection to attachment selection, the configuration state determines the corresponding product visual. If the customer changes the attachment, the interface can display the corresponding configured machine image, creating the feeling that the customer is actually building their machine rather than simply selecting a checkbox.

15. E-Commerce & Stripe Integration

The platform also incorporates payment functionality using **Stripe**, providing a secure payment layer for applicable purchasing workflows. The payment flow moves from product/configuration review through checkout, Stripe payment processing, and into the order/customer workflow. This allows the website to move beyond lead generation and support transactional customer journeys where required.

16. Customer Journey

The overall website journey was designed around a simple progression:

```
LANDING PAGE
      |
EXPLORE MACHINES
      |
SELECT MACHINE
      |
VIEW SPECIFICATIONS
      |
CONFIGURE MACHINE
      |
SELECT ATTACHMENTS
      |
CHECK COMPATIBILITY
      |
UPDATE MACHINE IMAGE
      |
REVIEW CONFIG
      |
ENQUIRY / PURCHASE
      |
STRIPE
```

This provides a clear path from **discovery** → **configuration** → **decision** → **conversion**.

17. Responsive Website Development

The platform was developed as a responsive web experience, designed to work across desktop, laptop, tablet, and mobile. Particular attention was given to configuration interfaces because complex selection interfaces can become difficult to use on smaller screens. The goal was to maintain the same business logic while adapting the UI to different screen sizes.

18. Lead Capture & Conversion

The website also contains customer enquiry and lead-generation functionality. Customers can move from product discovery into an enquiry journey without having to leave the product context, allowing product information and customer interest to remain connected. Lead information can then be stored and processed through the backend system.

19. Google Maps Integration

Google Maps API was integrated into relevant website/business workflows. The integration supports location-based experiences and provides useful location information for customers and operational workflows — particularly relevant to a business operating across Australia.

20. Meta Pixel Integration

Meta Pixel was implemented to track relevant website activity and support marketing and conversion measurement. The integration helps connect website interactions with digital marketing activities, allowing important customer actions to be measured across the website journey.

21. Content & Blog Experience

The website also includes content-focused functionality for business communication and SEO. Blog/content sections provide the business with a way to publish:

- Industry information
- Product-related content
- Company updates
- Educational content
- Search-oriented content

This complements the product experience and provides additional entry points for potential customers.

22. AI Chatbot Integration

The website also includes an AI-powered customer assistance experience. However, this was treated as **one supporting component of the overall website rather than the primary project**. The AI assistant helps customers with product questions, product discovery, recommendations, FAQs, customer enquiries, and machine information, and integrates with the product ecosystem to guide users toward relevant machinery.

> *Separate case study: AI-Powered Chatbot & AI Agent.*

23. Freight Integration

Freight management was also connected to the broader SYM ecosystem. The website/platform can connect machinery and customer requirements with freight workflows, including freight jobs, pickup/delivery information, freight companies, quotes, quote acceptance, and notifications. However, freight management is treated as a **separate system/module** rather than the primary focus of this website case study.

> *Separate case study: Freight Management System.*

24. Backend & API Architecture

The platform backend was developed using Node.js, Express.js, MongoDB, Mongoose, and REST APIs, providing the data and business logic required by the interactive frontend. Important backend responsibilities include:

- Product data
- Configuration data
- Compatibility rules
- Customer enquiries
- Payment-related workflows
- Integrations
- AI communication

- Business workflows

25. Production Infrastructure

The platform was deployed and maintained in a production cloud environment, configured to support reliable application hosting and API communication.

DigitalOcean → Ubuntu → Nginx → PM2 → Node.js → MongoDB

26. CI/CD & Deployment

GitHub Actions was used as part of the deployment workflow.

```
Developer
   |
  Git
   |
 GitHub
   |
GitHub Actions
   |
Production Deployment
   |
DigitalOcean
   |
Nginx + PM2
```

This provided a more consistent deployment process for production changes.

Technical Highlights

Interactive Product Configuration

Customers can configure machines instead of simply viewing static product pages.

Compatibility Engine

Machine and attachment relationships determine which options are available.

Dynamic Option Disablement

Invalid attachment combinations are automatically disabled.

Visual Configuration

Changing an attachment can dynamically change the machine's visual representation.

Stripe Integration

Secure payment processing was integrated into applicable customer purchasing workflows.

Full-Stack Architecture

The website combines React.js, Node.js, Express.js, MongoDB, and REST APIs.

Production Deployment

The platform runs within a real production infrastructure using DigitalOcean, Nginx, PM2, and CI/CD.

Key Engineering Challenge

The most interesting engineering challenge was making the **product configuration system behave like a real configurator rather than a collection of dropdowns**.

The system needed to answer questions such as:

> *"Can this attachment work with this machine?"*

and:

> *"If this attachment is selected, which other options should become unavailable?"*

The resulting logic can be summarized as:

```
Machine Selected
      |
Compatibility Matrix
      |
Available Options
      |
User Selects Attachment
      |
Recalculate Compatibility
      |
Disable Conflicting Options
      |
Update Configuration
      |
Update Product Image
```

This makes the configuration experience both **technically accurate and visually interactive**.

Result

The SYM Forklifts project transformed a traditional industrial machinery website into a more modern, interactive digital product experience. The platform combines:

Product Discovery + Interactive Configuration + Compatibility Rules + Dynamic Visuals + E-Commerce + Lead Generation + AI Assistance + Business Integrations

The key difference is that customers are not simply browsing a forklift catalogue.

They can **interactively configure a machine, understand which attachments are compatible, see the configuration change visually, and continue directly toward an enquiry or purchase.**

This approach creates a more engaging customer experience while also enforcing the business's actual product compatibility rules.

My Contribution

I contributed across the complete technical lifecycle of the platform, including:

- React.js frontend development
- Interactive product UI
- Product configuration experience
- Attachment selection system
- Compatibility logic
- Dynamic configuration state management
- Product image switching
- Node.js / Express backend
- MongoDB data modeling
- REST API development
- Stripe integration
- Google Maps integration
- Meta Pixel integration
- Lead-generation workflows
- AI integration
- Freight system integration
- Production deployment
- Nginx configuration
- PM2 process management
- DigitalOcean infrastructure
- GitHub Actions / CI/CD
- Production debugging and maintenance

Technology Stack

React.js · JavaScript · Node.js · Express.js · MongoDB · Mongoose · REST APIs · Stripe · Google Maps API · Meta Pixel · OpenAI · OpenClaw · AI Agents · DigitalOcean · Nginx · PM2 · GitHub Actions · CI/CD

Project Highlights

Interactive Product Configurator

Tesla-style machine configuration with dynamic attachment selection and visual updates.

Compatibility Engine

Automatically prevents incompatible machine and attachment combinations.

Dynamic Product Visualization

Machine imagery changes according to the customer's selected configuration.

E-Commerce Integration

Stripe-powered payment workflow for applicable purchasing journeys.

Full-Stack Production Platform

React + Node.js + Express + MongoDB deployed on production cloud infrastructure.

AI & Business Integrations

AI assistance, Google Maps, Meta Pixel, email/SMS services, and supporting business workflows.